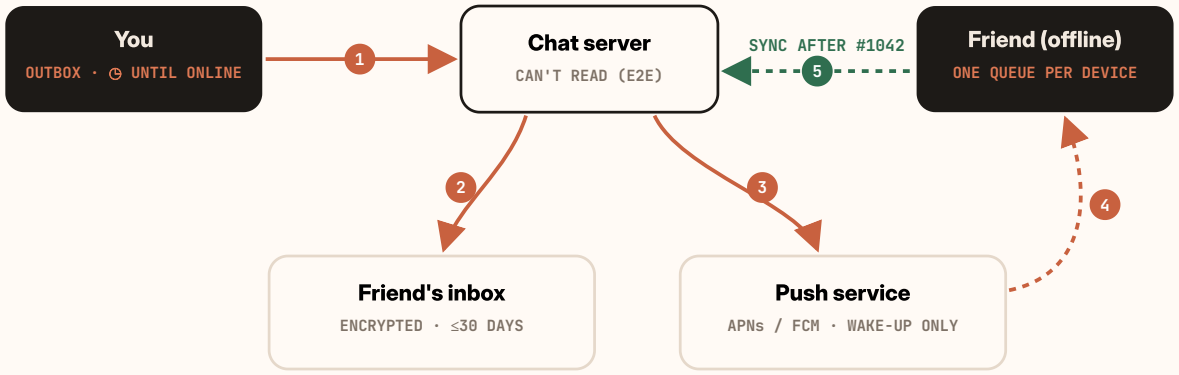
• SYSTEM DESIGN UNLOCKED · EP 04

Design WhatsApp: *Offline Messages*

Where messages wait when either phone is offline, and how 200 of them arrive in order.

THE FLOW



1 No signal? Message waits in your phone's outbox (◷), auto-sends later
2 Friend offline → server parks it in their inbox, encrypted, ≤30 days
3 Server asks Apple/Google push service to wake the phone
4 Push carries only a wake-up signal, never the message text
5 Phone reconnects: "everything after my last message ID"
6 Batches, oldest first; each receipt clears it from the inbox

WHERE A MESSAGE CAN WAIT

◷ **Your outbox.** You're offline. It hasn't left your phone; sends automatically when you reconnect.

✓ **Friend's inbox.** Server has it; friend is offline. Kept encrypted up to 30 days.

✓✓ **Delivered.** Friend's device pulled it and sent a receipt. Server deletes its copy.

REQUIREMENTS

- Never lose a message, either side offline
- Arrive in order after reconnect
- Wake sleeping phones without leaking content
- Survive a dead battery mid-download
- Work on every linked device independently

30 days MAX HOLD FOR UNDELIVERED

1 INBOX QUEUE PER DEVICE

0 MESSAGE TEXT IN THE PUSH (GOOD DESIGN)

EDGE CASES → FIX

Battery dies mid-download → nothing deleted without a receipt; resume from the last received ID.

50 chats flood in → fetch in batches; collapse into one summary notification.

Phone off, laptop on → each linked device has its own identity key + queue, so it still receives.

Offline past 30 days → undelivered messages are deleted; sender stays on ✓.

Send while offline → outbox keeps message IDs, so retries never create duplicates.

TRADE-OFFS TO SAY OUT LOUD

Wake-up push vs content push: private, but needs one extra fetch before showing the message.

Delete on delivery: less stored data, but no server history; backups live on the device.

Cursor sync (after last ID): cheap + ordered, but the client must track its cursor safely.

Per-device queues: devices work alone, but every message is stored and sent once per device.

INTERVIEW KEYWORDS TO SAY

`offline queue` `outbox` `store-and-forward` `per-device inbox` `push notification` `APNs / FCM` `wake-up signal` `cursor-based sync` `last message ID` `batching` `delivery receipt` `idempotency` `multi-device`

Sources: faq.whatsapp.com/665923838265756 · whatsapp.com/legal/privacy-policy · engineering.fb.com/2021/07/14/security/whatsapp-multi-device. Push payload, batching and cursor sync are standard interview design choices, not claims about WhatsApp's code.